Two-Stage Operational Amplifier with Miller Compensation

Project Overview

Designed and analyzed a two-stage operational amplifier using Miller compensation to improve stability and frequency response. Proper transistor sizing, biasing, and compensation capacitor selection were performed to meet targeted analog performance specifications.

Performance Specifications

Parameter	Achieved Value
DC Gain	60 dB
Phase Margin	66°
Unity Gain Bandwidth	39.8 MHz
Slew Rate	33.3 V/µs
-3 dB Bandwidth	39 kHz
Load Capacitance (CL)	3 pF

Simulation Analysis

DC Analysis: Verified biasing conditions and transistor operating regions.

AC Analysis: Evaluated gain, bandwidth, phase response, phase margin, and unity-gain bandwidth.

Transient Analysis: Examined large-signal response and slew-rate performance.

Miller Compensation

Miller compensation was employed to improve stability and frequency response. The compensation capacitor was selected as part of the design process to achieve the targeted analog performance.

Key Concepts

Two-stage op-amp design • Miller compensation • Analog VLSI • Transistor sizing and biasing • Frequency response • Stability and phase margin • AC analysis • Transient analysis • Slew-rate analysis • Circuit simulation

Results & Conclusion

Simulation results closely matched the expected theoretical behavior, confirming the correctness and reliability of the design. The project demonstrates practical understanding of analog VLSI design, compensation, frequency-response optimization, and circuit-level simulation.